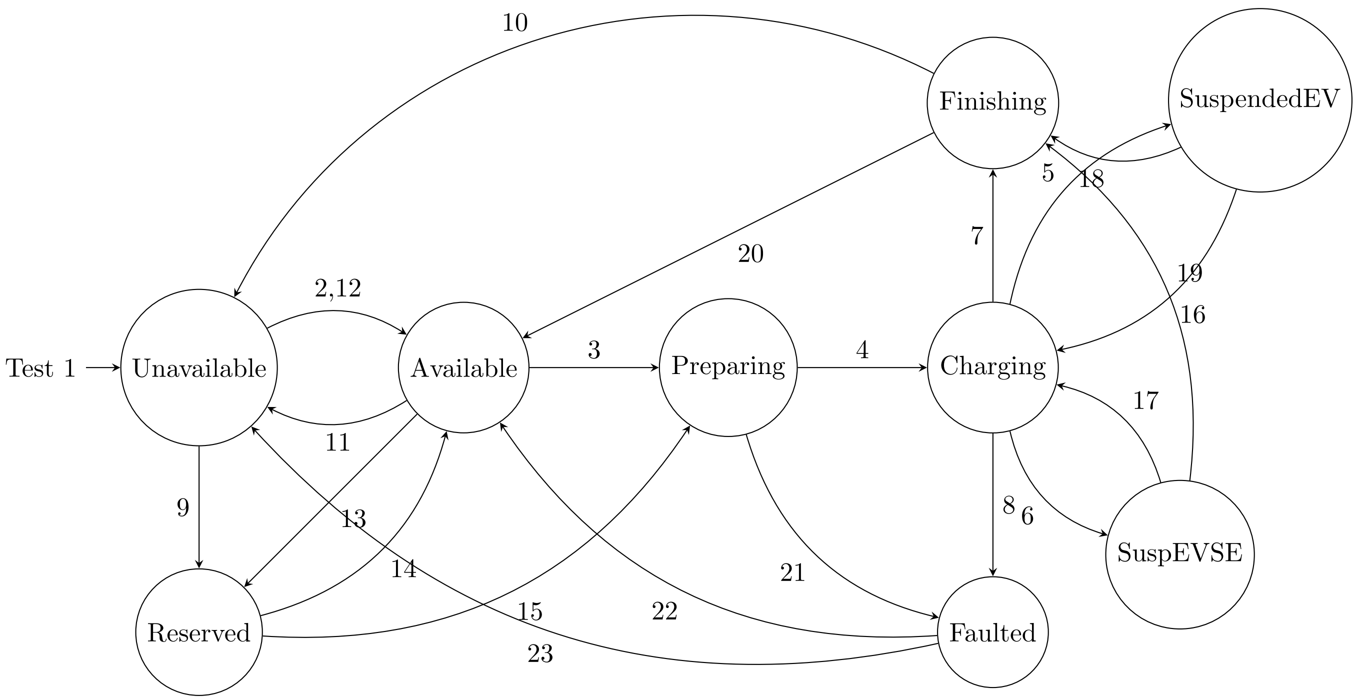
Evse.py Unit Test Report

MEA OCPP Certification (Automated)

January 15, 2026

State Transition Diagram



Unit Test Execution Results

ID	Test Case	State Check	Log Preview / Trace	Result
1	test_initial_state	Logic Check	`WHITE-beet-EI firmware version: 1.0.0` `RFIDSim stopped.`	PASSED

2	test_transition_to_available	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`Set the CP mode to EVSE`
			`Set the CP duty cycle to 100%`
			`Start the CP service`
			`Start SLAC in EVSE mode`
			`RFIDSim started. Call simulate_scan(id_tag) to trigger.`
			`State transition: Unavailable -> Available`
				PASSED

3	test_transition_to_preparing	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`RFIDSim started. Call simulate_scan(id_tag) to trigger.`
			`RFID Scanned: RFID_TAG`
			`EVSE Authorized by OCPP: RFID_TAG`
			`EV already connected, transitioning to Preparing for`
			`authorized tag: RFID_TAG`
			`State transition: Available -> Preparing`
			`EV already connected`
				PASSED

4	test_transition_to_charging	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`"Energy transfer mode selected" received`
			`State transition: Preparing -> Charging`
			`Maximum voltage: 400`
			`Maximum current: 32`
			`RFIDSim stopped.`
				PASSED

5	test_transition_to_suspended_ev	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`"DC Charge Parameters Changed" received`
			`EV maximum current: 32A`
			`EV maximum voltage: 400V`
			`EV ready: True`
			`Error code: 0`
			`SOC: 50%`
			`EV target current: 0A`
			`State transition: Charging -> SuspendedEV`
			`RFIDSim stopped.`
				PASSED

ID	Test Case	State Check	Log Preview / Trace	Result
6	test_transition_to_suspended_evse	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: Charging -> SuspendedEVSE RFIDSim stopped.	PASSED
7	test_transition_to_finishing	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "Session stopped" received State transition: Charging -> Finishing Closure type: 0 RFIDSim stopped.	PASSED
8	test_transition_to_faulted	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "Session Error" received State transition: Charging -> Faulted Session error: 0: Unspecified RFIDSim stopped.	PASSED
9	test_transition_to_reserved	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Unavailable -> Reserved Reservation set until 2026-01-01 12:00:00+00:00 RFIDSim stopped.	PASSED
10	test_transition_to_unavailable_via_pending	Logic Check	WHITE-beet-EI firmware version: 1.0.0 Scheduling availability change to Inoperative State transition: Finishing -> Available Pending 'Inoperative' availability detected, transitioning to Unavailable. State transition: Available -> Unavailable RFIDSim stopped.	PASSED
11	test_transition_available_to_unavailable	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Available -> Unavailable RFIDSim stopped.	PASSED

ID	Test Case	State Check	Log Preview / Trace	Result

12	test_transition_unavailable_to_available	Logic Check	`WHITE-beet-EI firmware version:  1.0.0`
			`State transition:  Unavailable -> Available`
			`RFIDSim stopped.`
			`============================== 23 passed in 0.04s`
			`==============================`
			`RFIDSim stopped.`
			`RFIDSim stopped.`
				PASSED

13	test_transition_available_to_reserved	Logic Check	`WHITE-beet-EI firmware version:  1.0.0`
			`State transition:  Available -> Reserved`
			`RFIDSim stopped.`
				PASSED

14	test_transition_reserved_to_available	Logic Check	`WHITE-beet-EI firmware version:  1.0.0`
			`State transition:  Reserved -> Available`
			`RFIDSim stopped.`
				PASSED

15	test_transition_reserved_to_preparing	Logic Check	`WHITE-beet-EI firmware version:  1.0.0`
			`EV already connected`
			`State transition:  Reserved -> Preparing`
			`RFIDSim stopped.`
				PASSED

16	test_transition_suspendedev_to_charging	Logic Check	`WHITE-beet-EI firmware version:  1.0.0`
			`"DC Charge Parameters Changed" received`
			`EV maximum current:  32A`
			`EV maximum voltage:  400V`
			`EV ready:  True`
			`Error code:  0`
			`SOC: 50%`
			`EV target current:  10A`
			`State transition:  SuspendedEV -> Charging`
			`RFIDSim stopped.`
				PASSED

ID	Test Case	State Check	Log Preview / Trace	Result

17	test_transition_suspendedevse_to_charging	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`"DC Charge Parameters Changed" received`
			`EV maximum current: 32A`
			`EV maximum voltage: 400V`
			`EV ready: True`
			`Error code: 0`
			`SOC: 50%`
			`EV target current: 10A`
			`State transition: SuspendedEVSE -> Charging`
			`RFIDSim stopped.`
				PASSED

18	test_transition_suspendedev_to_finishing	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`State transition: SuspendedEV -> Finishing`
			`RFIDSim stopped.`
				PASSED

19	test_transition_suspendedevse_to_finishing	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`State transition: SuspendedEVSE -> Finishing`
			`RFIDSim stopped.`
				PASSED

20	test_transition_finishing_to_available	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`State transition: Finishing -> Available`
			`RFIDSim stopped.`
				PASSED

21	test_transition_preparing_to_faulted	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`"Session Error" received`
			`State transition: Preparing -> Faulted`
			`Session error: 0: Unspecified`
			`RFIDSim stopped.`
				PASSED

22	test_transition_faulted_to_available	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`Resetting EVSE...`
			`State transition: Faulted -> Available`
			`RFIDSim stopped.`
				PASSED

23	test_transition_faulted_to_unavailable	Logic Check	`WHITE-beet-EI firmware version: 1.0.0`
			`State transition: Faulted -> Unavailable`
			`RFIDSim stopped.`
				PASSED

A Raw Unit Test Logs

```
============================= test session starts ==============================
platform darwin -- Python 3.14.0, pytest-9.0.2, pluggy-1.6.0 -- /Users/rus/Development/MEA-V2G/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/rus/Development/MEA-V2G
configfile: pytest.ini
collecting ... collected 23 items

tests/unit/test_evse_states.py::TestEvseStates::test_initial_state PASSED [  4%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_available_to_reserved PASSED [  8%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_available_to_unavailable PASSED [ 13%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_faulted_to_available PASSED [ 17%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_faulted_to_unavailable PASSED [ 21%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_finishing_to_available PASSED [ 26%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_preparing_to_faulted PASSED [ 30%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_reserved_to_available PASSED [ 34%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_reserved_to_preparing PASSED [ 39%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedev_to_charging PASSED [ 43%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedev_to_finishing PASSED [ 47%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedevse_to_charging PASSED [ 52%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedevse_to_finishing PASSED [ 56%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_available PASSED [ 60%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_charging PASSED [ 65%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_faulted PASSED [ 69%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_finishing PASSED [ 73%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_preparing PASSED [ 78%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_reserved PASSED [ 82%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_suspended_ev PASSED [ 86%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_suspended_evse PASSED [ 91%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_unavailable_via_pending PASSED [ 95%]
tests/unit/test_evse_states.py::TestEvseStates::test_transition_unavailable_to_available PASSED [100%]

==================================== PASSES ====================================
______________________ TestEvseStates.test_initial_state _______________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_____________ TestEvseStates.test_transition_available_to_reserved _____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: Available -> Reserved
```

```
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
___________ TestEvseStates.test_transition_available_to_unavailable ____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: Available -> Unavailable
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_____________ TestEvseStates.test_transition_faulted_to_available ______________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
Resetting EVSE...
State transition: Faulted -> Available
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
____________ TestEvseStates.test_transition_faulted_to_unavailable _____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: Faulted -> Unavailable
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
____________ TestEvseStates.test_transition_finishing_to_available _____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: Finishing -> Available
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_____________ TestEvseStates.test_transition_preparing_to_faulted ______________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"Session Error" received
State transition: Preparing -> Faulted
Session error: 0: Unspecified
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_____________ TestEvseStates.test_transition_reserved_to_available _____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: Reserved -> Available
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_____________ TestEvseStates.test_transition_reserved_to_preparing _____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
```

```
EV already connected
State transition: Reserved -> Preparing
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
____________ TestEvseStates.test_transition_suspendedev_to_charging ____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"DC Charge Parameters Changed" received
EV maximum current: 32A
EV maximum voltage: 400V
EV ready: True
Error code: 0
SOC: 50%
EV target current: 10A
State transition: SuspendedEV -> Charging
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
___________ TestEvseStates.test_transition_suspendedev_to_finishing ____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: SuspendedEV -> Finishing
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
___________ TestEvseStates.test_transition_suspendedevse_to_charging ___________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"DC Charge Parameters Changed" received
EV maximum current: 32A
EV maximum voltage: 400V
EV ready: True
Error code: 0
SOC: 50%
EV target current: 10A
State transition: SuspendedEVSE -> Charging
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
__________ TestEvseStates.test_transition_suspendedevse_to_finishing ___________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: SuspendedEVSE -> Finishing
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_________________ TestEvseStates.test_transition_to_available __________________
----------------------------- Captured stdout call -----------------------------
```

```
WHITE-beet-EI firmware version: 1.0.0
Set the CP mode to EVSE
Set the CP duty cycle to 100%
Start the CP service
Start SLAC in EVSE mode
RFIDSim started. Call simulate_scan(id_tag) to trigger.
State transition: Unavailable -> Available
__________________ TestEvseStates.test_transition_to_charging __________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"Energy transfer mode selected" received
State transition: Preparing -> Charging
Maximum voltage: 400
Maximum current: 32
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
__________________ TestEvseStates.test_transition_to_faulted ___________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"Session Error" received
State transition: Charging -> Faulted
Session error: 0: Unspecified
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_________________ TestEvseStates.test_transition_to_finishing __________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"Session stopped" received
State transition: Charging -> Finishing
Closure type: 0
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_________________ TestEvseStates.test_transition_to_preparing __________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
RFIDSim started. Call simulate_scan(id_tag) to trigger.
RFID Scanned: RFID_TAG
EVSE Authorized by OCPP: RFID_TAG
EV already connected, transitioning to Preparing for authorized tag: RFID_TAG
State transition: Available -> Preparing
EV already connected
__________________ TestEvseStates.test_transition_to_reserved __________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
```

```
State transition: Unavailable -> Reserved
Reservation set until 2026-01-01 12:00:00+00:00
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
________________ TestEvseStates.test_transition_to_suspended_ev ________________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"DC Charge Parameters Changed" received
EV maximum current: 32A
EV maximum voltage: 400V
EV ready: True
Error code: 0
SOC: 50%
EV target current: 0A
State transition: Charging -> SuspendedEV
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
_______________ TestEvseStates.test_transition_to_suspended_evse _______________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
"DC Charge Parameters Changed" received
EV maximum current: 32A
EV maximum voltage: 400V
EV ready: True
Error code: 0
SOC: 50%
EV target current: 10A
State transition: Charging -> SuspendedEVSE
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
__________ TestEvseStates.test_transition_to_unavailable_via_pending ___________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
Scheduling availability change to Inoperative
State transition: Finishing -> Available
Pending 'Inoperative' availability detected, transitioning to Unavailable.
State transition: Available -> Unavailable
--------------------------- Captured stdout teardown ---------------------------
RFIDSim stopped.
___________ TestEvseStates.test_transition_unavailable_to_available ____________
----------------------------- Captured stdout call -----------------------------
WHITE-beet-EI firmware version: 1.0.0
State transition: Unavailable -> Available
--------------------------- Captured stdout teardown ---------------------------
```

```
RFIDSim stopped.
============================== 23 passed in 0.04s ==============================
RFIDSim stopped.
RFIDSim stopped.
```